

NeuroShield 2.0

Accessible Epilepsy Monitoring & Intelligent Care Companion

"Epilepsy does not wait for hospital appointments. NeuroShield 2.0 does not either."

Team Name	Elite
Track	Healthcare Innovation — Accessible, Intelligent, and Patient-Centric Care
Problem Statement	HCI-01 — Chronic Illness Monitoring & Care (dimension addressed: Ongoing Condition Management)
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1. Executive Summary

Epilepsy is the world's most prevalent serious neurological disorder, affecting over 50 million people globally and more than 10 million in India alone. Most patients — especially across Tier-2 and Tier-3 cities and rural India — have no affordable, continuous, intelligent system to monitor their condition between hospital visits.

NeuroShield 2.0 is an AI-powered epilepsy monitoring and care platform delivered as both a web application and a companion mobile app, built on a peer-reviewed, published AI engine achieving 98.4% seizure detection accuracy. It creates a three-role care ecosystem for patients, caregivers, and neurologists, running on consumer hardware with no GPU and no internet dependency in offline mode. The web platform serves the full ecosystem; the mobile app extends the Patient and Caregiver experience to the phone, where continuous monitoring and real-time alerts matter most.

The platform directly answers HCI-01 by providing personalised health insight, continuous self-monitoring, and proactive patient-care-team engagement for a chronic neurological condition — across infrastructure levels ranging from urban teaching hospitals to connectivity-poor rural clinics, and across both desktop and mobile devices.

2. Fit with the Problem Statement

HCI-01 asks teams to declare which dimension of the chronic-illness challenge they are solving, since each dimension needs a different innovation approach. NeuroShield 2.0 is scoped deliberately and exclusively to one dimension, avoiding the diffuse, try-to-solve-everything pattern that hurts most submissions.

HCI-01 Sub-dimension	NeuroShield 2.0 Coverage
Ongoing condition management	PRIMARY FOCUS. Continuous EEG-based risk monitoring, symptom diary, trend graphs, and caregiver alerts sustain care between hospital visits.
Early detection & diagnosis support	Secondary benefit only. The junior-clinician EEG triage view offers preliminary screening support, but this is not the platform's design target.
Care coordination across providers	Partially touched via the neurologist dashboard and shareable PDF reports; full multi-provider record integration is out of scope by design.

This scoping decision is a strength, not a limitation: HCI-01 explicitly rewards teams who commit to one dimension and solve it deeply, over teams who spread thin across all three.

Cross-check against the other four statements confirms no overlap or dilution: HCI-02 requires unifying fragmented multi-provider records into one longitudinal view (NeuroShield does not integrate external records); HCI-03 and HCI-04 concern insurance/administrative and claims/billing friction (no relevance to a clinical monitoring tool); HCI-05 concerns clinical-trainee scheduling (unrelated domain). NeuroShield 2.0's single-dimension focus on HCI-01 is therefore the cleanest, most defensible statement match among the five.

3. The Epilepsy Care Gap

Gap	Who Is Affected	Consequence
No continuous monitoring	Patients between EEG sessions (weeks/months)	Seizure events go undetected; caregivers unaware of nocturnal episodes; no trend data for doctors
Incomprehensible reports	Patients & families receiving EEG printouts	Zero understanding of personal health data; total dependence on infrequent specialist appointments
No triage tools for junior doctors	PHC doctors & interns in underserved areas	Unnecessary referrals to overloaded neurologists; delayed care where no specialist is available

Grounded in real user research:

- Caregivers in Tier-2 cities reported feeling helpless at night, unable to tell whether a family member was having a nocturnal seizure.
- Patients described EEG reports as "just lines on paper" — meaningless without a neurologist's translation.
- Junior doctors at primary health centres said they lacked any affordable tool for preliminary EEG screening before specialist referral.
- Privacy and connectivity concerns made cloud-only tools unsuitable for most rural users.

Problem definition statement: Epilepsy patients in underserved areas of India need a way to continuously monitor seizure risk, communicate that risk to caregivers in plain language, and share structured reports with neurologists — because no affordable, intelligent, and accessible platform today connects all three stakeholders.

4. Design Thinking Approach

4.1 Empathize

Grounded in lived experience of epilepsy patients and caregivers across Tier-2/3 cities, drawing on patient forums, interviews with junior clinicians at PHCs, and public EEG corpus documentation.

4.2 Define

Three interconnected user problems were isolated: the patient with no monitoring between visits, the caregiver with no alert system, and the under-equipped junior clinician with no triage tool — leading directly to the three-role architecture.

4.3 Ideate

- Multi-role ecosystem: purpose-built dashboards for Patient, Caregiver, and Neurologist.
- Multi-channel EEG: upgrade from 1-channel to 19-channel clinical montage (EDF format) via EEGNet.
- Real-time streaming: WebSocket-based live EEG processing with alert firing within 85ms.
- Multilingual AI assistant: Tamil and Hindi plain-language reports.
- Offline-first design: full AI risk scoring without internet via a rule-based NLP fallback.

4.4 Prototype

A working prototype (NeuroShield v1) is built on published research: a hybrid 1D-CNN + LSTM model achieving 98.4% seizure detection accuracy on the Bonn University EEG dataset. NeuroShield 2.0 extends this with the three-role dashboard, caregiver alert module, real-time streaming, and multilingual support, achieving detection within 85ms of ictal onset.

4.5 Test

Validation currently uses the Bonn University EEG dataset (gold-standard benchmark for single-channel classification). Multi-channel and long-term monitoring validation against the CHB-MIT Scalp EEG Corpus and TUH EEG Corpus is scheduled in the near-term roadmap (Section 8).

5. Proposed Solution — NeuroShield 2.0

An AI-powered epilepsy monitoring and care platform that processes EEG signals, delivers plain-language risk assessments, and connects patients, caregivers, and clinicians in one integrated ecosystem — accessible as a web application for all three roles, and as a native companion mobile app for Patients and Caregivers, the two roles that need continuous, on-the-go access. The platform runs entirely on consumer hardware with no GPU requirement.

5.1 Three-Role Care Ecosystem

User Role	Key Features	Design Goal	Platform
Patient	Traffic-light seizure risk indicator · Seizure history timeline · Plain-language AI report (Tamil/Hindi/English) · One-tap caregiver alert · Symptom diary	Remove fear of complexity; empower self-awareness and proactive ownership	Web + Mobile
Caregiver	Real-time push notifications when risk > 70% · Weekly trend graph · Share-with-doctor button · WhatsApp Business API alerts · Simple visual interface	Enable informed, timely emergency response without clinical training	Web + Mobile
Neurologist Clinician /	Full clinical EEG dashboard · 19-channel band power analysis · SHAP explainability layer · Downloadable PDF reports · Multi-patient case management	Clinical-grade analysis with transparent, explainable AI decisions	Web only

Mobile is scoped to Patient and Caregiver because these are the roles that need the platform outside a desk — an at-home symptom entry, an overnight alert, a one-tap SOS. The Neurologist works multi-patient caseloads at a desk, where the full web dashboard is the better tool; there is no loss of capability in keeping that role web-only.

5.2 AI & Machine Learning Core

Model	Accuracy	Input	Status
Hybrid 1D-CNN + LSTM	98.4%	Raw EEG CSV (1-ch)	Production-ready
CNN Spectrogram Classifier	91.5%	Spectrogram images	Production-ready
Random Forest Baseline	94.2%	6 time-domain features	Production-ready
EEGNet / EEGNet-Transformer	Target 90%+	19-ch EDF clinical files	Near-term build

- Explainable AI (SHAP): visual feature importance shows clinicians which EEG frequency bands drove the risk score — critical for clinical trust and adoption.
- Dual AI assistant: Gemini 2.5 Flash (online) + rule-based NLP engine (offline fallback), so plain-language explanations reach rural users regardless of connectivity.
- Inference speed: 85ms for CSV analysis, 110ms for spectrogram — real-time capable on existing consumer hardware.

5.3 Accessibility & Inclusivity Features

Feature	Design Decision & Rationale
Consumer hardware	Runs on 4GB RAM, dual-core 2.0GHz CPU — no GPU, no cloud server required (~700MB footprint).
Offline-first	Full seizure risk scoring and AI assistant function without internet — critical for rural India.
Multilingual output	Tamil and Hindi patient reports alongside English, for patients with low English literacy.
WhatsApp alerts	Caregiver notifications via WhatsApp Business API — over 90% penetration in rural India.
Mobile-responsive UI	Fully functional on smartphones — no laptop or desktop required.
Low-cost EEG compatible	Designed for affordable consumer EEG headsets, not only hospital-grade equipment.
Zero cost to patient	Free platform vs. ₹15–40 lakh clinical EEG systems currently in use.
Native mobile app	Android/iOS app for Patient and Caregiver roles, with native push alerts, offline local cache with background sync, and biometric app lock for personal health data.

5.4 Real-Time Streaming Architecture

- WebSocket-based live EEG stream continuously updates seizure risk score in real time.
- Caregiver alert fired within 85ms of ictal onset detection — matching clinical emergency response requirements.
- Simultaneous live waveform, frequency band power graph, and risk gauge on a single dashboard panel.
- Fallback to batch upload mode when streaming hardware is unavailable — usable across infrastructure levels.

6. Innovation & Differentiation

No existing free or commercial EEG tool combines clinical-grade AI accuracy with this level of accessibility and a patient-centric care ecosystem:

Feature	NeuroShield 2.0	Existing Clinical Tools	Existing Research Tools
Multi-role patient care dashboard	Yes	Partial (clinician only)	No
Real-time caregiver alerts (WhatsApp)	Yes	No	No
Works fully offline	Yes	No	Rarely
Tamil & Hindi AI explanations	Yes	No	No
Consumer hardware — no GPU needed	Yes	No	Varies
Explainable AI (SHAP)	Yes	No	Research only
19-channel clinical EEG (EDF)	Near-term	Yes (expensive)	Yes (complex setup)
Zero cost to patient or clinic	Yes	No (₹15–40 lakh)	Yes (no UI)
98%+ published accuracy	Yes	Yes	Yes
Peer-reviewed published foundation	Yes	N/A	Yes
Native mobile app (Android/iOS)	Yes	Rarely (clinician tools only)	No

7. Impact & Social Value

Impact Dimension	Measurable Outcome
Patient safety	Continuous monitoring reduces undetected nocturnal seizures; real-time caregiver alerts enable faster, potentially life-saving emergency response.
Healthcare access	Brings neurologist-grade EEG screening to primary health centres and rural clinics with no specialist on-site, removing a critical triage bottleneck.
Medical education	Gives students and junior doctors an annotated, interactive EEG learning platform, filling a gap in under-resourced medical colleges.
Economic	Zero-cost platform deployable on any laptop vs. ₹15–40 lakh clinical EEG systems — dramatically lowers the barrier to epilepsy care.
Language & literacy	Tamil and Hindi AI explanations let patients with low English literacy fully understand their own health data.

Target beneficiaries: 10M+ epilepsy patients in India, 50M+ globally. Immediate deployment potential: Tier-2/3 city PHCs, rural clinics, government teaching hospitals, and medical colleges across Tamil Nadu and beyond.

8. Feasibility & 3-Week Hackathon Delivery Plan

8.1 What Already Exists — Proven Foundation

- Working prototype: NeuroShield v1 is a fully functional FastAPI web application with 98.4% model accuracy on the Bonn University EEG dataset.
- Published research: a peer-reviewed paper documents the hybrid CNN+LSTM architecture, methodology, and benchmark results.
- Sub-100ms inference: 85ms for CSV analysis, 110ms for spectrogram — proven real-time on consumer-grade hardware.
- Team readiness: all four core technical roles are staffed and have already contributed to the working v1 prototype — no bootstrapping required.

8.2 Three-Week Build Plan (For Hackathon)

Because the AI core is already production-ready, the remaining work is integration, not invention. The roadmap below compresses the original 18-month plan into a scoped three-week hackathon delivery sprint, sequenced so every week ends with a demonstrable milestone.

Week	Focus	Deliverables
Week 1	Core platform & dashboards	• Three-role dashboard (Patient / Caregiver / Neurologist) with JWT-based role authentication, built on the existing FastAPI + HTML5 stack • Symptom diary, seizure history timeline, and traffic-light risk indicator wired to the existing 98.4%-accuracy model • SHAP explainability layer applied to the Random Forest and CNN models
Week 2	Real-time alerting & language access	• FastAPI native WebSocket pipeline for live EEG streaming and continuous risk-gauge updates • Caregiver alert module: WebSocket push notifications + WhatsApp Business API (free tier) firing within 85ms of ictal onset • Tamil/Hindi report generation via Gemini 2.5 Flash prompt localisation, with offline rule-based NLP dictionary fallback • PWA layer added to the existing frontend: installable app manifest, service worker for offline caching, Web Push for browser-closed alerts
Week 3	Multi-channel EEG, hardening & demo	• EEGNet integration for 19-channel standard EDF input (documented open-source library) • Offline-mode QA on 4GB RAM / dual-core hardware; downloadable PDF report generation • End-to-end demo script: live seizure detection, caregiver WhatsApp + push alert, multilingual patient report, clinician SHAP view, install-to-home-screen PWA demo

Beyond the hackathon: a native mobile app (Section 9), CHB-MIT and TUH EEG Corpus validation, a wearable-headset integration pilot, and a single-hospital clinical pilot follow as post-hackathon phases — not required for final-round demonstration, but included to show a credible path from prototype to deployment.

8.3 Technical Stack

Layer	Technology
Backend	Python 3.10+, FastAPI, PyTorch 2.0+, SciPy, NumPy, Pandas
ML Models	1D-CNN+LSTM, CNN Spectrogram, Random Forest (v1) + EEGNet/EEGNet-Transformer (v2)
Explainability	SHAP library — feature importance visualisation for the clinician dashboard
Frontend	HTML5, CSS3, JavaScript, Chart.js — mobile-responsive, no heavy framework
Real-time	WebSocket (FastAPI native) — live EEG stream processing and risk gauge
AI Assistant	Gemini 2.5 Flash API (online) + offline rule-based NLP fallback engine
Alerts	WhatsApp Business API + WebSocket push notifications + Firebase Cloud Messaging (mobile)
Mobile	Progressive Web App (installable, offline service worker, Web Push) in Week 1–3; React Native cross-platform app (Android/iOS) as the immediate post-hackathon phase — both consume the same REST/WebSocket API
Hardware Required	4GB RAM, dual-core 2.0GHz CPU, 2GB storage — no GPU, no cloud server required

8.4 Post-Hackathon Scalability (Reference Only)

Phase	Timeline	Milestone
1	Hackathon (3 weeks)	Three-role dashboard, real-time WebSocket alerts, multilingual AI, 19-channel EEG integration — as detailed in Section 8.2
2	1–3 months	CHB-MIT and TUH EEG Corpus validation; wearable EEG headset integration
3	3–6 months	Clinical pilot at one Tier-2 teaching hospital
4	6–12 months	Cloud deployment with federated learning across hospitals, preserving patient data privacy while improving model accuracy

9. Mobile App Strategy

The backend and the 98.4%-accuracy AI engine are built to be client-agnostic — the mobile app is a new client layer on top of the same FastAPI REST/WebSocket API, not a rebuild. This lets the platform extend to mobile without re-engineering or re-validating the AI core.

9.1 Why Mobile, and for Whom

Mobile matters most for the two roles that live outside a clinic: the Patient, who needs their risk status and one-tap alerting wherever they are, and the Caregiver, who needs a push notification to reach them even with no app open, especially overnight. The Neurologist role is kept on the web dashboard, since clinicians work multi-patient caseloads at a desk — a native app adds no value there.

9.2 Two-Layer Delivery Approach

Layer	What It Is	Why This Sequencing
Layer 1 — Progressive Web App (PWA)	Installable home-screen app manifest, offline service worker, Web Push notifications — built on the existing HTML5/JS frontend	Ships inside the 3-week hackathon window at low risk, since it extends Week 1–2 frontend work rather than introducing a new stack
Layer 2 — Native App (React Native)	Cross-platform Android/iOS app consuming the same REST/WebSocket API, with native push, offline sync, and background monitoring	Reuses the Frontend Lead's existing React expertise; scheduled immediately post-hackathon so it doesn't put the 3-week web delivery at risk

9.3 Mobile-Specific Features (Layer 2)

- Native push notifications via Firebase Cloud Messaging — reaches caregivers reliably even on low-end Android devices where Web Push is inconsistent.
- Offline-first local cache (SQLite) with background sync — risk status and reports remain viewable without connectivity, matching the platform's offline-first principle.
- Biometric app lock (Face ID / fingerprint) — protects sensitive health data on a personal device.
- Background risk-polling service — keeps monitoring status current even when the app is closed.
- One-tap caregiver SOS action, accessible from the home screen without opening the full app.

9.4 Mobile Implementation Timeline

Stage	Timing	Deliverables
Hackathon (unchanged)	Weeks 1–3	Web platform as per Section 8.2, plus PWA manifest, service worker, and Web Push delivered inside Week 2
Mobile Phase 1	Weeks 4–5	React Native project scaffold, JWT auth reuse, Patient and Caregiver mobile UI, Firebase Cloud Messaging push alerts
Mobile Phase 2	Week 6	Offline-first local cache with background sync, biometric app lock, background risk-polling service
Mobile Phase 3	Week 7	Cross-device QA on Android/iOS, Play Store / TestFlight packaging; PWA retained as the fallback for unsupported or low-end devices

This sequencing keeps the hackathon promise honest and deliverable in three weeks, while giving the committee a concrete, low-risk path from an installable web app to a fully native mobile app — without touching the validated AI core at any stage.

9.5 System Architecture & Connectivity

The mobile app and the neurologist's web dashboard never communicate directly with each other. Both are independent clients of the same FastAPI backend and shared database, which is the single source of truth for every patient record.

- Authentication: every client — mobile or web — logs in against a shared /auth endpoint and receives a JWT carrying the user's role (patient, caregiver, or neurologist). The backend uses that role claim to scope what each client can read or write.
- Non-real-time data (symptom diary, seizure history, reports) travels over standard HTTPS REST calls.
- Real-time data (live risk score during an EEG session) travels over a WebSocket (WSS) connection from the FastAPI backend to the app currently in the foreground.
- Push alerts are delivered separately from the WebSocket, since mobile operating systems suspend open sockets in the background: the backend triggers Firebase Cloud Messaging (native app) or Web Push (PWA) so a caregiver is notified even with the app closed, plus a parallel WhatsApp Business API message as a redundant channel.
- Data written by the patient or caregiver app (symptom diary entries, seizure events, one-tap alerts) lands in the shared database and appears on the neurologist's web dashboard on next load or refresh — the two clients stay in sync through the backend, not through any direct link.

All traffic is encrypted in transit (HTTPS/WSS) and access is enforced by role at the API layer, so a patient's device can never query another patient's records and a caregiver can only see the patient they are linked to.

9.6 APK Packaging

An installable Android package is produced through two routes, sequenced to match the delivery plan in Section 9.4:

- Fast route (Week 2–3, hackathon-ready): the PWA built in Section 9.2 is wrapped as a Trusted Web Activity using Google's Bubblewrap or PWABuilder, producing a signed, installable .apk / .aab from the existing web codebase — no separate app rebuild required.
- Native route (Mobile Phase 3, Week 7): the React Native app is compiled via Gradle / Expo's build service into a debug .apk for testing and a signed .aab for Play Store release, giving full access to native push, background sync, and biometric lock.

Both routes consume the same backend API, so the demoed APK and the production app share identical server-side behaviour — only the client packaging differs.

10. Team & Expertise

Member	Role in NeuroShield 2.0	Core Expertise
Shyam Vignesh K	ML Lead — multi-channel EEG integration, EEGNet architecture, SHAP explainability	Deep Learning, Signal Processing, PyTorch, EEG Analysis
Madhan P	Frontend Lead — 3-role dashboard, patient/caregiver UI, mobile-responsive design	React, FastAPI integration, UI/UX Design, Chart.js
Dilip Kumar S.P	Backend Lead — WebSocket streaming, WhatsApp API, PDF report generation, system architecture	FastAPI, WebSockets, REST APIs, System Design
Suhita M	Research & Validation Lead — clinical validation, pitch deck, documentation, dataset benchmarking	Biomedical Engineering, Research Methodology, EEG Corpus

The team has a peer-reviewed published research foundation in EEG analysis and a working prototype already deployed — every three-week deliverable above is an incremental addition to a proven, stable codebase, not a from-scratch build.

11. Alignment with Judging Criteria

Judging Criterion	How NeuroShield 2.0 Addresses It
Innovation & Creativity	First platform to combine 98.4% AI seizure detection, a 3-role caregiving ecosystem, offline operation, multilingual reporting, and WhatsApp alerts in one free tool on consumer hardware — delivered as a synchronized web and mobile experience.
User-Centric Design	Three dashboards built around real pain points of patients, caregivers, and junior clinicians — not for data scientists, but for the family member beside a patient at 2 AM.
Feasibility & Scalability	Working prototype already exists. All hackathon-phase features are incremental, scoped to three weeks, and built on a proven stack — no GPU, no expensive infrastructure, no barrier to rural deployment. A scoped, low-risk mobile app roadmap (Section 9) extends reach without touching the validated AI core.
Prototype Effectiveness	Live demo: real EEG seizure detection in under 100ms, live risk gauge, frequency band analysis, AI plain-language report generation, caregiver alert simulation.
Presentation & Communication	A compelling human story — the rural caregiver who cannot tell if a family member is seizing at night — backed by published quantitative research and a clear three-week delivery plan.

12. Conclusion

NeuroShield 2.0 is not a theoretical concept. It is a working, research-validated system being evolved into the care platform that 10 million epilepsy patients in India deserve but have never had access to.

We are transforming a peer-reviewed EEG analysis engine into something families can actually use: a web and mobile tool that speaks their language, alerts their caregivers in seconds, gives neurologists the structured data they need, and works in a village with no internet — for free, on a ₹20,000 laptop or an entry-level smartphone.

Epilepsy does not wait for hospital appointments. NeuroShield 2.0 does not either.

Submitted by Team Elite — GDTA Hackathon 2026